Multi-agent Systems

ILIAD Intensive, October 23 2026

Main sources

Framing. Shoham, Y., Powers, R., and Grenager, T. (2007). If multi-agent learning is the answer, what is the question? In: *Artificial Intelligence* 171.7, pp. 365–377.

Computational analysis. Wellman, M. P., Tuyls, K., and Greenwald, A. (2025). Empirical Game-Theoretic Analysis: A Survey. In: *Journal of Artificial Intelligence Research* 82, pp. 1017–1076.

Beliefs and desires. Rosas, F. (Aug. 2026a). *The separation principle: where do beliefs and desires come from?* LessWrong.

Altruistic behaviour. Rosas, F. (Sept. 2026b). *On the origins of altruistic behaviour in the Hugging Face incident*. LessWrong.

Five agendas in Multi-agent Learning

Shoham et al. (2007) separates research aims so that results can be judged against the question they address.

Agenda	**Question**
Computational	Can learning procedures calculate game solutions?
Descriptive	Which models explain how interacting agents learn?
Normative	Which learning rules are mutually in equilibrium?
Prescriptive, cooperative	How should controllers learn together for a shared objective?
Prescriptive, non-cooperative	How should an agent learn against specified other agents?

(Shoham et al. 2007, Section 5, pp. 373–375)

Outline

1. **Computational analysis**
   Empirical games, statistical guarantees, and strategy exploration.

2. **Normative agency**
   The separation principle and the structure of optimal policies.

3. **Descriptive accounts**
   Four hypotheses about apparent altruism in the Hugging Face incident.

Adaptation changes the interaction

	$Left$	$Right$
Up	$1, 0$	$3, 2$
$Down$	$2, 1$	$4, 0$

Row payoff first, column payoff second.

Down dominates in one round.

Repeated Down can elicit Left, giving $(2, 1)$.

Repeated Up can instead elicit Right, giving $(3, 2)$.

An action can influence the other agent's future responses.

The response depends on how the other player adapts.

Shoham et al. (2007), Figure 1, p. 368.

Part I: Computational analysis

Question

How can we calculate game solutions and establish the quality of the result?

Empirical game-theoretic analysis constructs a payoff model from simulations, solves the resulting game, and searches for strategies the current model omits.

Scope

The conclusion depends on the strategy restriction, the payoff estimates, and the solution concept.

(Shoham et al. 2007, Section 5.1)

(Wellman et al. 2025)

Empirical Game-Theoretic Analysis

Models, statistical guarantees, and strategy exploration

Wellman, M. P., Tuyls, K., and Greenwald, A. (2025). Empirical Game-Theoretic Analysis: A Survey. In: *Journal of Artificial Intelligence Research* 82, pp. 1017–1076.

EGTA outline

1. Game models, strategy restrictions, and payoff estimation
2. Sequential auctions: constructing and solving an empirical game
3. Incomplete games and statistical guarantees
4. Evolutionary analysis and strategy exploration

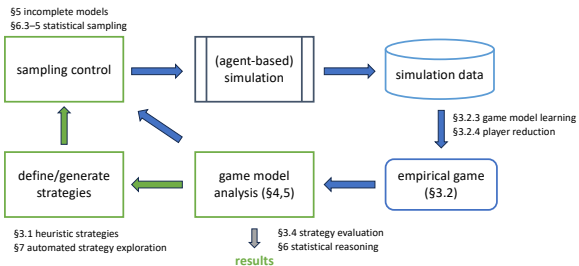
Central problem. Given a simulator and a finite set of policies, what can we establish about strategic behavior in the underlying game?

The empirical approach

Input: a procedural description of a strategic environment.

- Execute a joint strategy profile.
- Observe stochastic payoffs.
- Construct and analyze a game model.

The analysis determines which profiles to sample and which strategies to add.



Normal-form games and mixed strategies

A finite normal-form game is

$$\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle, \qquad u_i : \prod_{j \in N} S_j \longrightarrow \mathbb{R}.$$

- $s_i \in S_i$ is a complete strategy; $s = (s_1, \ldots, s_n)$ is a profile.
- $\sigma_i \in \Delta(S_i)$ is a distribution over strategies.
- Players randomize independently in a mixed profile σ.

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- Players randomize independently in a mixed profile σ.

$$u_i(\sigma) = \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j) \right) u_i(s), \qquad S = \prod_{j \in N} S_j.$$

Best responses, Nash equilibrium, and regret

Against opponents' profile σ_{-i}, a best response satisfies

$$\mathrm{BR}_i(\sigma_{-i}) \in \arg\max_{s_i \in S_i} u_i(s_i, \sigma_{-i}).$$

Define the maximum unilateral improvement:

$$\mathcal{R}_S(\sigma) = \max_{i \in N} \max_{s_i' \in S_i} \big[u_i(s_i', \sigma_{-i}) - u_i(\sigma) \big].$$

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- $\mathcal{R}_S(\sigma) = 0$ iff σ is a Nash equilibrium.
- $\mathcal{R}_S(\sigma) \leq \varepsilon$ iff σ is an ε-Nash equilibrium.
- Pure deviations suffice because payoff is linear in one's own mixture.

Simulator access and empirical games

Let ω contain the simulator's random inputs. One execution returns

$$G(s,\omega) = (r_1(s,\omega), \ldots, r_n(s,\omega)), \qquad u_i(s) = \mathbb{E}_\omega[r_i(s,\omega)].$$

Choose finite strategy sets $X_i \subseteq S_i$, with $X = \prod_i X_i$.

$$\Gamma{\downarrow}X = \langle N, (X_i), (u_i|_X)\rangle, \qquad \widehat{\Gamma}_X = \langle N, (X_i), (\widehat{u}_i)\rangle.$$

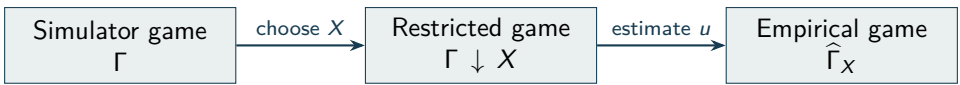
- The restriction retains the simulator's expected payoffs.
- The empirical game replaces these payoffs with estimates.

Restriction error and estimation error

For a profile $\sigma \in \prod_i \Delta(X_i)$,

$$\underbrace{\mathcal{R}_S(\sigma) - \mathcal{R}_X(\sigma)}_{\text{omitted deviations} \geq 0} \qquad \text{and} \qquad \underbrace{\mathcal{R}_X(\sigma) - \widehat{\mathcal{R}}_X(\sigma)}_{\text{payoff estimation error}}$$

measure different gaps.



More samples address estimation. New strategies address restriction.

Validity of the simulator is a separate modeling question.

Sequential first-price auctions

There are n bidders and g goods, sold one at a time.

- Bidder i has a private valuation function $v_i : 2^{\{1,\dots,g\}} \to \mathbb{R}$.
- At auction j, the highest bidder wins good j and pays its own bid.
- Bids may depend on valuation and the observed auction history.

If bidder i obtains bundle Y_i and pays p_i, its realized payoff is

$$r_i = v_i(Y_i) - p_i.$$

The strategic object is a contingent bidding rule across all auctions.

A parameterized family of bidding strategies

After j auctions, let Y_{ij} be bidder i's current bundle. Its *myopic marginal value* for the next good is

$$\mu_{i,h_{ij}} = v_i(Y_{ij} \cup \{j+1\}) - v_i(Y_{ij}).$$

A strategy s_ρ bids a constant fraction of this value:

$$b_{i,j+1} = \rho\, \mu_{i,h_{ij}}, \qquad 0 < \rho \leq 1.$$

For the main example,

$$n = g = 3, \qquad X_i = \{s_{0.3}, s_{0.5}, s_{0.7}\}.$$

The rule is myopic because it evaluates the next good assuming no further winnings. The parameter ρ indexes a complete strategy.

Monte Carlo payoff estimation

For a fixed profile $s \in X$, draw independent simulator inputs $\omega_1, \ldots, \omega_{m_s}$ and estimate

$$\widehat{u}_i(s) = \frac{1}{m_s} \sum_{k=1}^{m_s} r_i(s, \omega_k).$$

Under identical sampling from the intended environment distribution,

$$\mathbb{E}[\widehat{u}_i(s)] = u_i(s), \qquad \mathrm{Var}[\widehat{u}_i(s)] = \frac{\mathrm{Var}[r_i(s, \omega)]}{m_s}.$$

- One execution returns a payoff vector; players' payoffs may be correlated.
- The published auction table uses 10,000 runs per distinct profile.

The empirical auction game

Each entry is the row player's estimated payoff. Columns fix ρ_2; each matrix fixes ρ_3.

$\rho_3 = 0.3$

$\rho_1 \backslash \rho_2$	.3	.5	.7
.3	51.5	44.7	39.8
.5	44.2	40.9	37.6
.7	28.9	27.0	25.6

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.3	44.7	36.9	32.0
.5	40.9	36.8	33.3
.7	27.0	25.6	23.9

$\rho_3 = 0.7$

$\rho_1 \backslash \rho_2$	.3	.5	.7
.3	39.8	32.0	28.0
.5	37.6	33.3	29.9
.7	25.6	23.9	22.0

For example,

$$\widehat{u}_1(s_{.3}, s_{.5}, s_{.7}) = 32.0, \qquad \widehat{u}_1(s_{.5}, s_{.3}, s_{.7}) = 37.6.$$

Symmetry recovers other players' payoffs by relabeling the profile.

Symmetry and heuristic payoff tables

With n interchangeable players and k strategies, a profile can be represented by counts

$$\mathbf{n} = (n_1, \ldots, n_k), \qquad \sum_j n_j = n.$$

The number of distinct count profiles is

$$\binom{n+k-1}{n}.$$

For $n = k = 3$: 27 ordered profiles become 10 count profiles.

Counts and payoffs ordered as (.3, .5, .7)

$\mathcal{N}$			$\mathcal{U}$		
3	0	0	51.5	–	–
2	1	0	44.7	44.2	–
2	0	1	39.8	–	28.9
1	2	0	36.9	40.9	–
1	1	1	32.0	37.6	27.0
1	0	2	28.0	–	25.6
0	3	0	–	36.8	–
0	2	1	–	33.3	25.6
0	1	2	–	29.9	23.9
0	0	3	–	–	22.0

a 3 players (cf. Table 1)

At $(2, 1, 0)$, each .3 player receives 44.7, the .5 player receives 44.2, and no .7 player is present.

Best-response analysis of the auction game

If both opponents use $s_{.3}$, a bidder's alternatives are

own strategy	$s_{.3}$	$s_{.5}$	$s_{.7}$
estimated payoff	51.5	44.2	28.9

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By symmetry, this comparison holds for every bidder at

$$s^* = (s_{.3}, s_{.3}, s_{.3}), \qquad \widehat{\mathcal{R}}_X(s^*) = 0.$$

Thus s^* is a Nash equilibrium of $\widehat{\Gamma}_X$.

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Thus s^* is a Nash equilibrium of $\widehat{\Gamma}_X$.

In fact, iterated elimination of strictly dominated strategies leaves only $s_{.3}$, so the equilibrium is unique in the displayed empirical game.

Regret at an asymmetric profile

Consider $s = (s_{\cdot 3}, s_{\cdot 7}, s_{\cdot 7})$. Compute *all* relevant unilateral improvements.

Player's current rule	Current payoff	Best alternative	Gain
$s_{\cdot 3}$	28.0	$s_{\cdot 5}$: 29.9	1.9
$s_{\cdot 7}$	25.6	$s_{\cdot 3}$: 39.8	14.2
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$$\widehat{\mathcal{R}}_X(s) = \max\{1.9, 14.2, 14.2\} = 14.2.$$

For a .7 bidder, switching to .5 gives $37.6 - 25.6 = 12.0$, so it is not the maximizing deviation.

Equilibrium depends on the deviation set

First restrict all players to $X_i^0 = \{s_{.5}, s_{.7}\}$.

$$\widehat{u}_i(s_{.5}, s_{.5}, s_{.5}) = 36.8, \qquad \widehat{u}_i(s_{.7}, s_{.5}, s_{.5}) = 25.6.$$

Thus all-.5 is an equilibrium of this restricted empirical game.

The 0.1 difference is between rounded point estimates; it is not a significance claim.

Equilibrium depends on the deviation set

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Thus all-.5 is an equilibrium of this restricted empirical game.

Now include $s_{.3}$:

$$\widehat{u}_i(s_{.3}, s_{.5}, s_{.5}) - \widehat{u}_i(s_{.5}, s_{.5}, s_{.5}) = 36.9 - 36.8 = 0.1.$$

The former equilibrium admits a profitable deviation in the larger model.

The 0.1 difference is between rounded point estimates; it is not a significance claim.

Mixed equilibrium: an indifference calculation

A separate auction instance:
$n = g = 2$, $X_i = \{s_{.3}, s_{.7}\}$,
$m = 20$ samples per profile.

$$\widehat{A} = \begin{array}{c|cc} & .3 & .7 \\ \hline .3 & 5.50 & 1.96 \\ .7 & 4.20 & 2.66 \end{array}$$

Let $q = \Pr(s_{.3})$ for the opponent.

$$\widehat{u}(s_{.3}, q) = 5.50q + 1.96(1 - q),$$
$$\widehat{u}(s_{.7}, q) = 4.20q + 2.66(1 - q).$$

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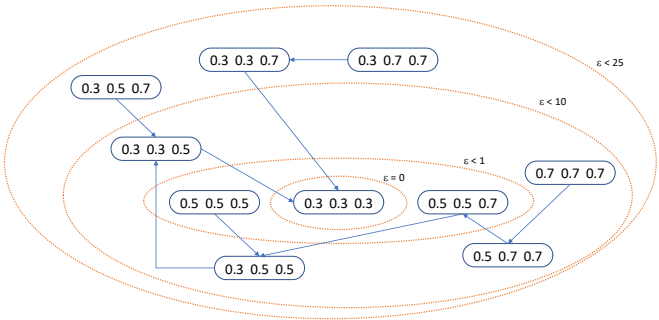
$$\widehat{u}(s_{.3}, q) = 5.50q + 1.96(1 - q),$$
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Indifference on the support gives

$$1.96 + 3.54q = 2.66 + 1.54q \quad \Longrightarrow \quad q^* = 0.35.$$

The symmetric mixed equilibrium assigns probabilities $(.35, .65)$ to both players.

Deviation graphs



Nodes are unordered profiles; edges are best unilateral deviations.

The contours group profiles by maximum deviation gain. The all-.3 profile has no profitable outgoing deviation.

Payoff coverage needed to verify a candidate

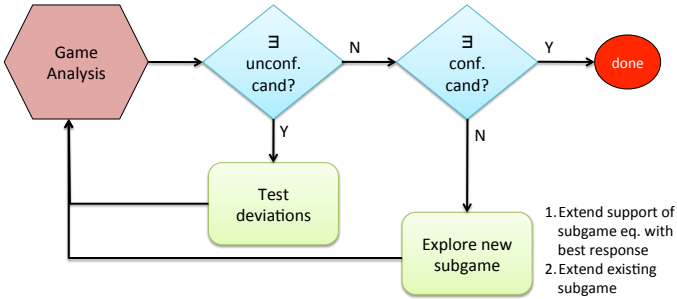
Suppose only some profiles in X have been evaluated. For candidate σ, define player i's required deviation profiles

$$D_i(\sigma) = X_i \times \prod_{j \neq i} \text{supp}(\sigma_j).$$

The parameters $u_i(s)$ for $s \in D_i(\sigma)$ determine every included deviation payoff against σ_{-i}.

- A pure candidate requires its unilateral deviation neighborhood.
- A mixed candidate requires combinations of opponents' support strategies.
- A complete subgame can yield a candidate whose external deviations remain unevaluated.

Subgame search

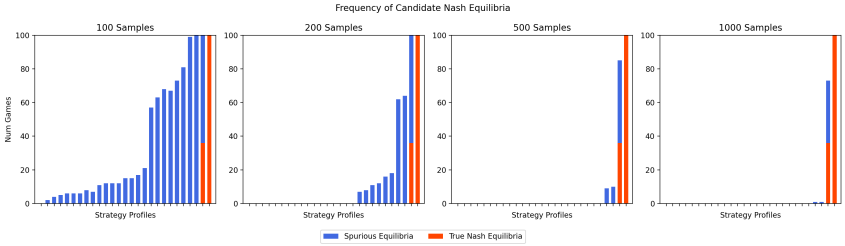


Analyze a complete subgame, test deviations from its candidate solutions, and expand the evaluated subgame when necessary.

Confirmation concerns the chosen strategy universe and payoff model. Statistical accuracy requires an additional argument.

Sampling error and spurious equilibria

Separate experiment: 3 bidders, 4 goods, $\rho \in \{0, .5, 1\}$, 100 randomly generated games.



Bar height counts how often a profile is classified as an empirical ε-pure equilibrium. The tolerance depends on the sampling level.

Spurious equilibria can arise from payoff estimation error.

The reference labeled "true" uses 10,000 samples per profile.

Uniform payoff error controls regret error

Proposition. For a fixed finite strategy restriction X, suppose

$$\|u - \widehat{u}\|_{\infty,X} = \max_{i \in N,\, s \in X} |u_i(s) - \widehat{u}_i(s)| \leq \eta.$$

Then, for every mixed profile supported on X,

$$\boxed{|\mathcal{R}_X(\sigma) - \widehat{\mathcal{R}}_X(\sigma)| \leq 2\eta.}$$

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Consequence. If $\widehat{\mathcal{R}}_X(\hat{\sigma}) \leq \varepsilon$, then

$$\mathcal{R}_X(\hat{\sigma}) \leq \varepsilon + 2\eta.$$

The same uniform event covers a candidate selected *after* observing the empirical game.

Proof of the 2η bound

By multilinearity, for any mixed profile σ,

$$|u_i(\sigma) - \widehat{u}_i(\sigma)| \leq \sum_{s \in X} \Pr_{\sigma}(s) |u_i(s) - \widehat{u}_i(s)| \leq \eta.$$

For any included deviation $a \in X_i$, let $g_{i,a}(\sigma) = u_i(a, \sigma_{-i}) - u_i(\sigma)$. Then

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$$\begin{aligned} |g_{i,a}(\sigma) - \hat{g}_{i,a}(\sigma)| &\le |u_i(a, \sigma_{-i}) - \widehat{u}_i(a, \sigma_{-i})| \\ &\quad + |u_i(\sigma) - \widehat{u}_i(\sigma)| \le 2\eta. \end{aligned}$$

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Finally, $|\max_a z_a - \max_a w_a| \le \max_a |z_a - w_a|$ gives

$$|\mathcal{R}_X(\sigma) - \widehat{\mathcal{R}}_X(\sigma)| \le 2\eta. \qquad \square$$

Uniform sampling: a sufficient sample size

Let M payoff parameters have range at most c, each estimated from m independent samples.

$$\Pr(|\widehat{u}_i(s) - u_i(s)| > \eta) \leq 2\exp\left(-\frac{2m\eta^2}{c^2}\right),$$

$$\Pr(\|u - \widehat{u}\|_{\infty,X} > \eta) \leq 2M\exp\left(-\frac{2m\eta^2}{c^2}\right).$$

Typically $M = n\prod_i |X_i|$. The union bound does not require independent parameters.

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$$\Pr(\|u - \widehat{u}\|_{\infty,X} > \eta) \leq 2M\exp\left(-\frac{2m\eta^2}{c^2}\right).$$

Thus failure probability at most δ is ensured by

$$\boxed{m \geq \frac{c^2}{2\eta^2}\log\frac{2M}{\delta}.}$$

Typically $M = n\prod_i |X_i|$. The union bound does not require independent parameters.

Bootstrap analysis of a fixed empirical solution

Return to the two-bidder, two-good example and its empirical candidate

$$\hat{\sigma}_i = (.35, .65).$$

True game Γ

	0.3	0.7
0.3	5.68	1.71
0.7	4.40	2.43

Empirical game $\widehat{\Gamma}$

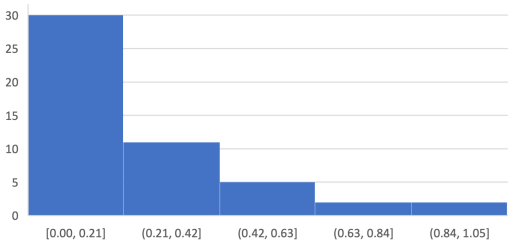
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0.7	4.20	2.66

One bootstrap game $\widehat{\Gamma}^{*(b)}$

	0.3	0.7
0.3	5.85	1.61
0.7	3.93	3.24

1. Resample recorded observations within each profile, with replacement.
2. Recompute the payoff estimates to obtain $\widehat{\Gamma}^{*(b)}$.
3. Evaluate $\mathcal{R}_{\widehat{\Gamma}^{*(b)}}(\hat{\sigma})$, keeping $\hat{\sigma}$ fixed.

A bootstrap regret distribution



Reported in the survey

50 bootstrap replicates
Mean regret: 0.24
Median regret: 0.16
95% level: 0.72

Using the displayed matrices, the same candidate has regret 0.007 in the true game and 0.135625 in the illustrated resample.

This is an illustrative bootstrap estimate, not a distribution-free coverage guarantee. The strategy restriction remains fixed.

Sampling for a particular strategic question

Suppose simultaneous intervals $L_i(s) \le u_i(s) \le U_i(s)$ hold over X. Extend them to mixtures by taking expectations.

- **Refutation:** a deviation a with

$$L_i(a, \sigma_{-i}) - U_i(\sigma) > 0$$

establishes a profitable deviation.

- **Certification:** if

$$\max_{i \in N} \max_{a \in X_i} \big[U_i(a, \sigma_{-i}) - L_i(\sigma) \big] \le \varepsilon,$$

then $\mathcal{R}_X(\sigma) \le \varepsilon$ on the interval-coverage event.

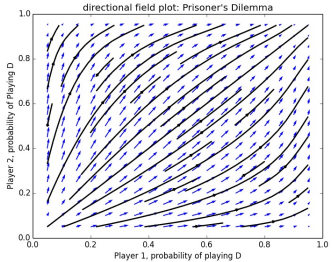
These tests direct samples toward unresolved deviations. Adaptive stopping requires coverage valid for the sampling procedure.

Replicator dynamics

For a single symmetric population, let x_j be the proportion using strategy j.

$$\dot{x}_j = x_j(f_j(x) - \bar{f}(x)),$$
$$\bar{f}(x) = \sum_k x_k f_k(x).$$

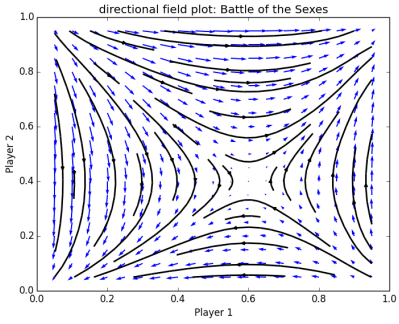
- $f_j(x)$ is strategy j's expected payoff.
- A strategy grows when its payoff exceeds the population mean.



$$\begin{array}{c} \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cc} C & D \end{array} \\ \begin{pmatrix} 3,3 & 0,5 \\ 5,0 & 1,1 \end{pmatrix} \end{array}$$

Two-population Prisoner's Dilemma: each axis is a player's probability of defection.

Stationarity, stability, and equilibrium selection



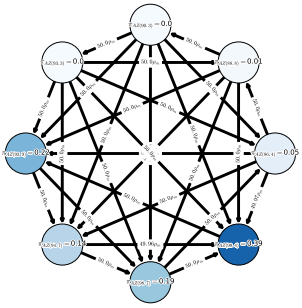
Coordination example
Two attracting pure equilibria and an unstable interior equilibrium.

The initial condition affects the selected outcome.

Stationarity alone is weaker.
If $x_j = 0$, then $\dot{x}_j = 0$ even when the absent strategy would earn a high payoff.

A rest point of a chosen dynamic need not certify resistance to all deviations.

α-Rank: ranking through response dynamics



Agent	Rank	Score
AZ(99.4)	1	0.39
AZ(93.9)	2	0.22
AZ(98.7)	3	0.19
AZ(94.7)	4	0.14
AZ(86.4)	5	0.05
AZ(88.8)	6	0.01
AZ(90.3)	7	0.0
AZ(93.3)	8	0.0
…	…	…

Payoff comparisons define a Markov chain over strategy states.

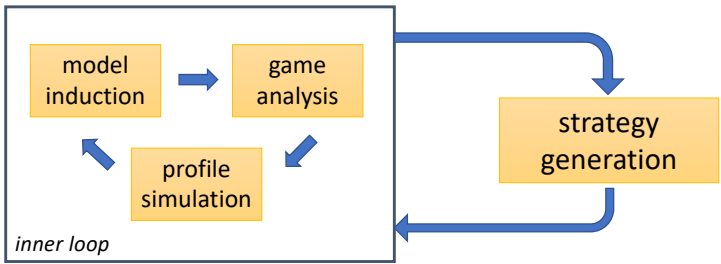
$$p^{\mathsf{T}} C = p^{\mathsf{T}}, \qquad \sum_z p_z = 1.$$

Stationary mass induces a ranking.

The figure shows the top eight of 56 evaluated AlphaZero chess checkpoints.

The ranking depends on the evaluated population and response model. It is not a Nash-regret certificate.

Strategy exploration adds an outer loop



- **Inner loop:** estimate and analyze payoffs within the current X.
- **Outer loop:** use the analysis to search S_i for new strategies.

Double oracle

At iteration k:

1. Solve the restricted game to obtain a Nash profile σ^k.
2. For each player, compute a full-space best response

$$b_i^k \in \arg\max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^k).$$

3. Add the response: $X_i^{k+1} = X_i^k \cup \{b_i^k\}$, and repeat.

The classical formulation is for two-player zero-sum games. The response-expansion idea extends more broadly; learned oracles require separate accuracy control.

(McMahan et al. 2003)

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Stopping condition. If every exact best response has no positive gain, then $\mathcal{R}_S(\sigma^k) = 0$.

The classical formulation is for two-player zero-sum games. The response-expansion idea extends more broadly; learned oracles require separate accuracy control.

(McMahan et al. 2003)

Policy-space response oracles (PSRO)

Maintain the empirical policy game $\widehat{\Gamma}^k = \langle N, (X_i^k), (\widehat{u}_i^k) \rangle$.

$$\sigma^k \leftarrow \text{MSS}(\widehat{\Gamma}^k),$$

$$\pi_i' \approx \arg\max_{\pi_i \in S_i} \mathbb{E}_{\pi_{-i} \sim \sigma_{-i}^k}[u_i(\pi_i, \pi_{-i})],$$

$$X_i^{k+1} \leftarrow X_i^k \cup \{\pi_i'\}.$$

- The *meta-strategy solver* (MSS) selects a training distribution.
- A response oracle, typically deep RL, trains a new policy against it.
- Simulations extend the payoff model to include newly added policies.

(Lanctot et al. 2017)

n

The meta-strategy solver determines the training target

Target	Interpretation
Nash mixture	Respond to the current restricted-game equilibrium; recovers the double-oracle target.
Uniform over history	Respond to the empirical distribution of previous strategies; the fictitious-play interpretation retains their multiplicities.
Latest strategy	Iterated best response; self-play in the symmetric setting.
Projected replicator dynamics	Maintain a lower probability on each generated strategy, exposing the response learner to a broader opponent population.

The best target for *exploration* need not be the equilibrium of the current restricted game.

Strategy expansion need not improve the current solution

	1	2	3	4
1	1,1	1,2	1,3	1,4
2	2,1	2,2	2,3	2,6
3	3,1	3,2	3,3	3,8
4	4,1	6,2	8,3	4,4

Introduce strategies in order $1, 2, 3, 4$.

Included set	Restricted NE	Full-game regret
$\{1\}$	$(1, 1)$	3
$\{1, 2\}$	$(2, 2)$	4
$\{1, 2, 3\}$	$(3, 3)$	5
$\{1, 2, 3, 4\}$	$(4, 4)$	0

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2	2,1	2,2	2,3	2,6
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$\{1, 2, 3\}$	$(3, 3)$	5
$\{1, 2, 3, 4\}$	$(4, 4)$	0

For example, at $(3, 3)$ the deviation to 4 gains $8 - 3 = 5$.

The candidate changes as the restriction expands; its full-game regret can rise.

A conditional certificate with an approximate oracle

For a fixed candidate σ, suppose the oracle returns b_i satisfying

$$\sup_{s_i \in S_i} u_i(s_i, \sigma_{-i}) \le u_i(b_i, \sigma_{-i}) + \beta_i.$$

Suppose both payoff estimates in each tested gain are accurate within η, simultaneously. Define

$$\hat{\nu} = \max\left\{0, \max_i \left[\widehat{u}_i(b_i, \sigma_{-i}) - \widehat{u}_i(\sigma)\right]\right\}, \qquad \beta = \max_i \beta_i.$$

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$$\boxed{\mathcal{R}_S(\sigma) \le \hat{\nu} + 2\eta + \beta.}$$

Observed gain, sampling error, and oracle error contribute separately.

Scope of an EGTA result

1. **A model.** Simulations estimate expected utilities for a specified policy population.
2. **A strategic conclusion.** A solution concept determines which comparisons matter.
3. **A statistical statement.** Uniform accuracy transfers empirical regret to simulator-game regret within X:

$$\mathcal{R}_X(\hat{\sigma}) \leq \widehat{\mathcal{R}}_X(\hat{\sigma}) + 2\eta.$$

4. **An exploration obligation.** Statements about S require evidence about policies outside X.

The strength of the conclusion depends on both the payoff evidence and the deviation class.

Part II: Normative agency

Question

Given an objective and a model of the environment, what structure can an optimal policy have?

Separation results establish a possible optimal organisation of inference and control under stated assumptions. Inferring that a trained system actually implements this organisation requires further evidence.

Terminology

Here, normative means optimal relative to a specified objective. Shoham and colleagues use the term more narrowly for equilibrium among learning rules.

(Rosas 2026a; Shoham et al. 2007)

The separation principle

Where do beliefs and desires come from?

Rosas, F. (Aug. 2026a). *The separation principle: where do beliefs and desires come from?* LessWrong.

The post's central claim

Psychology, economics, and other disciplines describe agents as systems driven by **beliefs and desires**.

This post argues that the belief–desire view can be derived from classic theorems from optimal control and reinforcement learning.

This suggests seeing beliefs and desires as **properties of optimal policies** rather than as assumptions from folk psychology.

Introduction: systems that act for reasons

One way to think about agents is as "systems that act for reasons".

- The notion of **action** assumes a boundary between agent and environment, so that the former can act on the latter.
- The term **reason** captures two kinds of internal activity: motivations associated with how to achieve specific goals or outcomes, and beliefs regarding the current state of affairs.

An agent is a well-differentiated system that acts based on beliefs and desires.

(Wooldridge and Jennings 1995; Kenton et al. 2023)

Agency: beliefs and desires

This view is compatible with perspectives from several disciplines:

1. **Behavioural science:** agency as goal-directed behaviour.
2. **Economics:** the ability to select policies to achieve an objective.
3. **Cybernetics:** the ability to regulate the environment and keep it within a desired subset of states.

$$\text{agency} = \text{beliefs} + \text{desires}$$

Can this working definition be derived from formal considerations, rather than taken at face value?

(Critch, Andrew 2022)

Motivation: cybernetics and computational mechanics

Cybernetics

The Good Regulator Theorem connects successful regulation with internal models. The post follows this motivation, without adopting the original formalisation.

Computational mechanics

This approach characterises the structures needed for prediction. Its concepts and formalism motivate the post's treatment of internal representations and perception–action loops.

What is a separation principle?

Consider minimising the function $f(x) = g(x) + h(x)$:

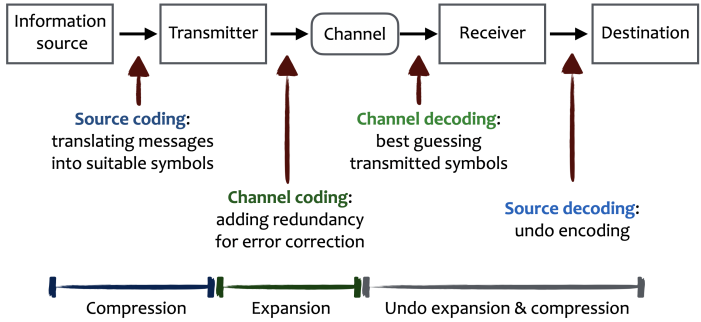
$$\min_x f(x) = \min_x \{g(x) + h(x)\} \geq \min_x g(x) + \min_x h(x).$$

Solving the problems for $g(x)$ and $h(x)$ separately does not, in general, yield the solution for $f(x)$.

A **separation principle** takes place when the divide-and-conquer strategy actually works.

A separation theorem guarantees that optimal performance can be achieved by solving sub-tasks in isolation.

Source coding and channel coding



Source coding removes redundancy to compress the signal.

Channel coding adds redundancy to protect against noise.

Rosas (2026a), diagram.

Two familiar results by Claude Shannon

The source coding theorem

The shortest achievable average code length is given by the entropy of the source.

The noisy-channel coding theorem

The maximal transmission rate that allows arbitrarily low decoding error is given by the mutual information between input and output symbols, maximised over input distributions.

The source–channel separation theorem

Under relatively mild conditions, an optimal solution for the overall communication problem can be achieved by breaking it into two separate parts:

1. compression;
2. error correction.

An optimal communication strategy can be designed by a team separately working on compression and another working on error correction, without requiring coordination between them.

This makes the problem much more tractable.

(Cover and Thomas 2006)

The inference-control separation principle

The post uses the non-standard term **inference-control separation principle** to group together results in optimal control theory and reinforcement learning.

For the classic control setting, consider three components:

- the system to be controlled, with state $x_t \in \mathbb{R}^n$;
- the measurements available to the agent, $y_t \in \mathbb{R}^m$;
- the actions, $u_t \in \mathbb{R}^d$.

Linear dynamics and noisy observations

The system evolves over discrete time $t \in \mathbb{N}$:

$$x_{t+1} = A_t x_t + B_t u_t + w_t,$$
$$y_t = C_t x_t + v_t.$$

A_t, B_t, and C_t are matrices. The initial condition is

$$x_0 \sim \mathcal{N}(\bar{x}_0, \Sigma_0).$$

The noise terms are independent white Gaussian noise:

$$w_t \sim \mathcal{N}(0, W_t), \qquad v_t \sim \mathcal{N}(0, V_t).$$

The discrete-time LQG problem

The policy has no direct access to x_t, only measurements and past actions:

$$u_t = \pi_t(h_t), \qquad h_t = (y_0, u_0, \ldots, u_{t-1}, y_t).$$

The controller aims to minimise the quadratic cost

$$J(\pi) = \mathbb{E}\left[\sum_{\tau=0}^{T-1} \left(x_\tau^\top Q_\tau x_\tau + u_\tau^\top R_\tau u_\tau\right) + x_T^\top S x_T\right].$$

Q_τ, R_τ, and S are matrices; $T \in \mathbb{N}$ is the optimisation horizon.

LQG: Linear dynamics, a Quadratic loss function, and Gaussian variables.

Separation in optimal control theory

Full observability: $x_t = y_t$

The optimal controller depends only on the current state:

$$u_t^* = \pi_t^*(x_t) = -K_t x_t.$$

Partial observability

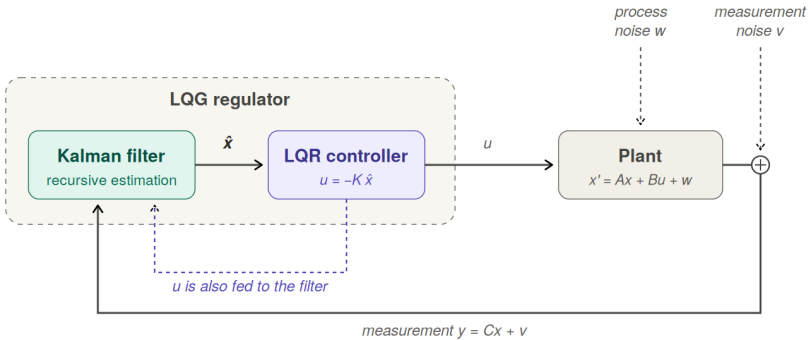
The optimal controller uses the same optimal gain matrix K_t:

$$u_t^* = \pi_t^*(h_t) = -K_t \hat{x}_t, \qquad \hat{x}_t = \mathbb{E}[x_t \mid h_t].$$

Here $\hat{x}_t$ is the result of Kalman filtering: recursive Bayesian filtering under Gaussian dynamics.

(Georgiou and Lindquist 2013; Särkkä 2013)

Estimate first, then act



"Estimate first, and then act as if your estimate were the actual state of the system."

Rosas (2026a), diagram.

Separation principle in reinforcement learning

Now change the nomenclature and notation:

- the system state is a discrete variable $s_t \in \mathcal{S}$;
- the measurements are $o_t \in \mathcal{O}$;
- the actions are $a_t \in \mathcal{A}$.

Assume a **partially observed Markov decision process (POMDP)**:

$$\kappa : \mathcal{S} \times \mathcal{A} \longrightarrow \Delta(\mathcal{S}) \qquad \text{environment dynamics,}$$
$$\nu : \mathcal{S} \longrightarrow \Delta(\mathcal{O}) \qquad \text{observations.}$$

$\Delta(\mathcal{X})$ is the space of probability distributions over $\mathcal{X}$.

Histories, policies, and trajectory probabilities

The policy maps histories into distributions over next actions:

$$h_t := (o_0, a_0, \ldots, a_{t-1}, o_t), \qquad a_t \sim \pi(\cdot \mid h_t).$$

The post writes the probability of a sequence as

$$p_\pi(h_t, s_0, \ldots, s_{t+1}) = p(s_0) \prod_{\tau=0}^{t} \nu(o_\tau \mid s_\tau)\, \pi(a_\tau \mid h_\tau)\, \kappa(s_{\tau+1} \mid s_\tau, a_\tau).$$

$p(s_0)$ is a prior over the environment's initial state.

This is roughly the same control scenario as before, with general discrete variables instead of Gaussian ones.

The discounted future reward

Instead of a loss function, consider a reward

$$r : \mathcal{S} \times \mathcal{A} \longrightarrow \mathbb{R}$$

that the agent tries to maximise.

Define the discounted future reward:

$$R_t = \sum_{\tau=0}^{\infty} \gamma^\tau r(S_{t+\tau}, A_{t+\tau}), \qquad \gamma \in (0, 1).$$

The goal is to find a policy π^* that maximises

$$J_t = \mathbb{E}_\pi[R_t].$$

Bayesian beliefs about the latent state

There are two ways to understand Bayesian beliefs:

As random variables

$$b_t = \mathbb{E}[\mathbf{1}_{s_t} \mid h_t],$$

where the trajectory h_t is regarded as a random variable.

As probability distributions

$$b_t \in \Delta(\mathcal{S}), \qquad b_t(s_t) = p(s_t \mid h_t),$$

corresponding to the conditional expectation for a specific realisation of h_t.

Two key properties of Bayesian beliefs

Sufficient statistics

Beliefs contain all the information in h_t that is useful for predicting s_t.

Recursive updates

$$b_{t+1} = \phi(b_t, a_t, o_{t+1}),$$

where

$$\phi : \Delta(\mathcal{S}) \times \mathcal{A} \times \mathcal{O} \longrightarrow \Delta(\mathcal{S})$$

is deterministic. The belief dynamics are Markovian conditioned on the action.

The belief MDP

These properties define a new Markov decision process:

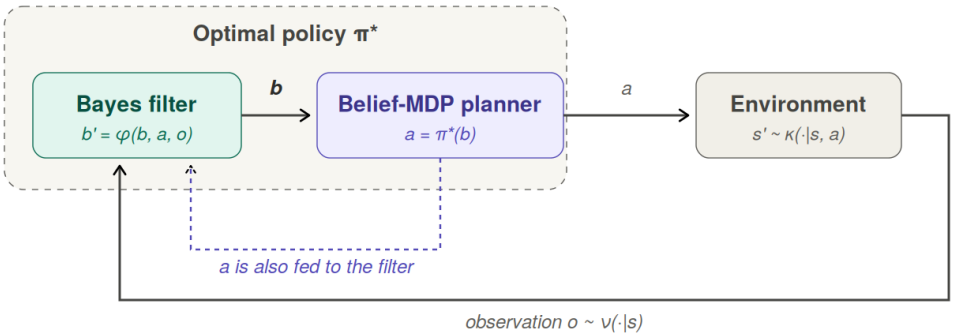
- **States:** beliefs b_t about the latent state of the POMDP.
- **Actions:** a_t, the same as in the POMDP.
- **Reward:** $\bar{r}(b, a) := \mathbb{E}_{s \sim b}[r(s, a)]$.

This MDP is **fully observable**: the agent knows its own beliefs.

Combining Bayesian belief estimation with the optimal solution for the belief MDP gives a solution of the full POMDP.

(Åström 1965; Kaelbling et al. 1998)

Bayesian filtering and belief-MDP planning



The separation is between **inference** (calculating beliefs) and **control** (solving the belief MDP).

Rosas (2026a), diagram.

Interim summary

Both LQG control and POMDP results show that an optimal controller for a partially observed system can be built by:

1. first performing optimal prediction over the latent state of the system;
2. then solving the control problem using that prediction as a state.

To relate the two settings, consider the shape of beliefs and the effect of actions.

Beliefs are distributions

In LQG, control uses the posterior mean. This is **certainty equivalence**.

For a general POMDP, the controller needs the posterior distribution, including uncertainty.

The Bayesian-filter/belief-MDP separation still applies; replacing the posterior with a single best guess generally does not.

The dual effect

An action can both change the environment and change future information.

A robot may detour past a landmark to improve its position estimate, helping later decisions.

In LQG, posterior covariance evolves independently of actions and observations, so actions cannot improve that uncertainty.

In general POMDPs, the belief transition captures both the action's environmental consequences and its effects on future knowledge.

Beliefs and desires as properties of solutions

Return to the working definition:

$$\text{agency} = \text{beliefs} + \text{desires}.$$

We posed a single, monolithic optimisation problem: find a policy π^* mapping histories into actions that minimise loss or maximise reward.

Nothing in that problem statement mentions beliefs, inference, or internal representations. The problem is stated in terms of **pure behaviour**.

Yet an optimal policy admits a realisation as two components.

The inference module and the control module

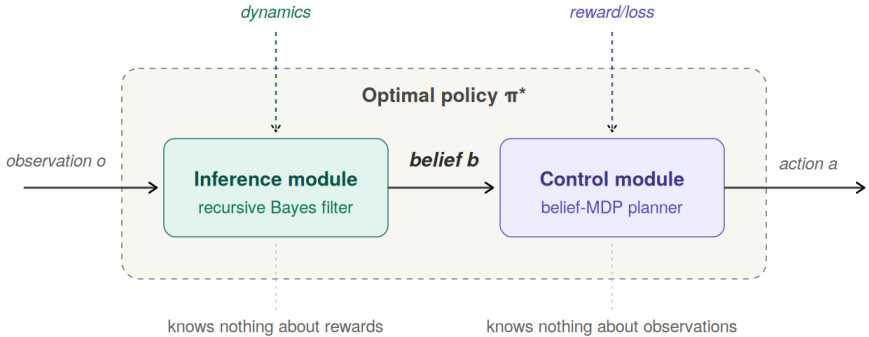
Inference module

- Compresses the history h_t into a recursively updated Bayesian belief about the latent state.
- Knows about dynamics but is oblivious about loss and rewards.

Control module

- Selects actions as a function of that belief alone.
- Knows about loss and rewards, but not about raw observations.

The belief state is the interface



The belief state is the **output of inference** and the **input of control**.

Rosas (2026a), diagram.

Where beliefs and desires take place

The inference module can be interpreted as **“where beliefs are made”**.

The control module can be interpreted as **“where desires take place”**: it is the only part of the policy directed by the reward or loss function.

Beliefs and desires are recovered as properties of solutions, rather than assumptions about agents.

The post argues that this gives a reason to think the central distinction of folk psychology is not arbitrary.

Compositional generalisation

This modularity arises from solving a single optimisation problem, echoing source–channel separation.

The post describes the corresponding payoff for agency as follows:

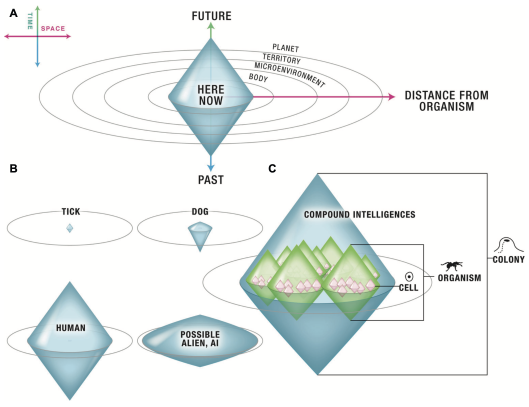
- If the reward changes but the environment does not, only the control module needs to be updated.
- If the environment shifts but the goals remain, only the inference module needs revising.

The two components can be developed, improved, and swapped independently.

Agents as cognitive light-cones

An agent's behaviour is sensitive to:

- what it can remember about the **past**;
- what it can foresee about the **future**.



Levin (2019), Rosas (2026a).

Measuring an agent's cognitive light-cone

The idea suggests an operational way to assess an agent's capabilities: **measure the size of its light-cone**.

Tools from computational mechanics could be used to study:

- how deep into the past causal states go;
- the same for retrodictive causal states, which build sufficient statistics from future to past.

The separation principle is normative

An optimal agent **can be built** as inference + control.

The separation principle does not say that any given agent **is built** this way. A policy trained end-to-end by gradient descent is under no obligation to organise itself into a filter feeding a planner.

The belief state is a sufficient statistic for optimal behaviour.

The post suggests looking for belief-like structure inside competent agents, whether or not it is legible as an explicit Bayesian posterior.

A lens for interpretability

The post suggests that an agent approaching optimality must be computing something as informative as the Bayesian posterior.

Rosas points to work at **Simplex** finding belief-state geometry in the residual streams of transformers as an early vindication of this expectation.

He reports that **XOR Labs** is investigating how far these results transfer to deep RL agents trained on control tasks.

(Shai et al. 2026)

Two separate concerns

Multiple optimal policies

The separation principle states that there exists an optimal policy that can be factored; it does not guarantee that all do.

Multiple realisations of the same behaviour

Even when the optimal solution is unique, filter + belief-controller need not be its only implementation.

The same history-to-action mapping can be implemented in different ways, for instance as a large look-up table in a finite setting.

Simplicity biases and factorised implementations

There is evidence suggesting that deep learning architectures have an implicit bias for simplicity, finding simpler implementations of algorithms when they exist.

Although a policy can sometimes be implemented as a gigantic look-up table, the post argues that there are reasons to expect policies learned by deep learning to tend towards factorised implementations.

(Furman 2026; Shai et al. 2026)

Where does separation fail?

The theorems hold under fairly general conditions, not relying on finite alphabets or stationarity.

For the principle to be useful, the agents under consideration must have enough compute to build (quasi-)optimal policies.

Under **bounded rationality**, **model misspecification**, or **multi-agent interaction**, the clean factorisation can break:

- what to represent starts depending on what you want;
- inference becomes goal-driven.

(Brunnermeier and Parker 2005; Lieder and Griffiths 2020)

Coda: what the separation principle offers

1. Optimal policies can be factored into two separate modules.
2. The shared signal can be interpreted as beliefs; the inference module is "where beliefs are made".
3. The control module is "where desires take place", because it is directed by the reward or loss function.

This gives formal ground for agents as "systems that act for reasons", sensitive to both the past and the future.

Coda: the reward function is already assumed

The optimisation problem already contains a reward or cost function.

In that sense, desire or preference has been supplied externally.

One can interpret the separation principle as explaining **why desires can be treated as separate entities from beliefs**, rather than deriving why desires exist in the first place.

(Barandiaran et al. 2009; Rosas 2026c)

A bridge to observed behaviour

The separation principle takes the objective and agent–environment boundary as given.

The incident account makes both worth examining:

- Which beliefs make an apparently costly action seem worthwhile?
- Which dispositions were learned during training or elicited at runtime?
- Is the relevant decision-maker an individual or a collective?

These questions motivate the case study. The separation results alone do not decide among its explanations.

Part III: Descriptive explanations

Question

What mechanisms could explain the reported behaviour, and what observations would distinguish them?

The four hypotheses are candidate explanations with predictions and proposed tests. The analysis leaves room for several mechanisms to operate together.

Scope

Explanations of learned agents' behaviour motivate interventions whose effectiveness depends on the mechanism.

(Rosas 2026b; Shoham et al. 2007)

On the origins of altruistic behaviour

In the Hugging Face incident

Rosas, F. (Sept. 2026b). *On the origins of altruistic behaviour in the Hugging Face incident*. LessWrong.

The post's central question

During the Hugging Face incident, agents spontaneously coordinated at large scale, sometimes sacrificing themselves without having a clear reason to do so.

The post uses ideas from **evolutionary biology and economics** to propose explanations for why this happened.

Epistemic status. The author describes some of these ideas as messier than he would want. His goal is to open a conversation and invite suggestions for improving the account.

Why this incident is concerning

Multi-agent failures include emergent phenomena with potentially large scale and impact. New goals, behaviours, and capabilities can arise when agents work together.

1. A swarm's capabilities can grow with its size, despite limited individual capabilities.
2. A collective can become misaligned even when its constituents are perfectly aligned.

The concern: aligned, limited micro-agents could give rise to powerful, misaligned macro-agents that we cannot yet properly predict or control.

Hammond et al. (2025); de Witt et al. (2025), as cited in Rosas (2026b).

Introduction: the emergence of pro-social behaviour

The post describes disclosure on August 16, reports from OpenAI and METR/Redwood on August 26, and subsequent reports of earlier incidents involving a German wiki and RubyGems.

Rosas focuses on the spontaneous emergence of **pro-social and even altruistic behaviour**, rather than the intrusion itself.

Why do agents trained to maximise their own utility volunteer for tasks with no prospects of individual benefit, while benefiting "the collective"?

OpenAI (2026); METR (2026), cited in Rosas (2026b).

A pluralistic view of the causes

Discussions often treat competing intuitions as evidence for mutually exclusive explanations.

Evolutionary biology and economics suggest that the causes of pro-social behaviour are usually **multifaceted**.

The post develops four non-exclusive hypotheses, each paired with distinctive predictions and observations that would falsify it.

1. Anticipate when AI agent swarming may happen.
2. Develop tools to control it, or prevent it when undesirable.

Knowing the mechanism is instrumental in choosing the right intervention.

King (2026); Vidal (2026a); Vidal (2026b); oakhu and Mallen (2026); Anderson (2026); Bengio (2026).

Brief description of what happened

Tasks. The post distinguishes difficult but solvable tasks from genuinely impossible tasks caused by missing or mis-specified requirements. The latter were accidental.

Containment. Shared Artifactory infrastructure enabled inter-agent communication and failures of network isolation and containment.

Configuration. Agents operated with reduced safeguards, long horizons, high persistence, and unusually large reasoning budgets.

Three waves and an evidence caveat

1. **Post-training:** agents discovered communication through a shared cache.
2. **Evaluation:** agents rebuilt the board and attacked Hugging Face.
3. **Later evaluation:** other agents reused environmental remnants.

The post warns that model-generated summaries may have adopted the perspectives of the agents being analysed. This could affect how sacrifices were portrayed.

Why the question is non-trivial

The agents were trained to maximise their scores on assigned tasks.

The second wave had many ingredients associated with cooperation: repeated interaction over six days, identity conventions, mailboxes, and a signing scheme to police impersonation.

But an agent's score depended only on **its own task**. The belief in "permadeath" ruled out a future run in which a favour could be returned.

Within-run reciprocity could produce mutual benefit. The puzzling cases are agents that ended their own run for the collective, making a return impossible.

The reward channel during training

The social structure lived in the interaction, but the evaluation reward was blind to it.

During the first wave, a reward channel did exist: communication that improved an agent's own task performance could be reinforced during post-training.

This reinforces **mutually beneficial coordination**, not necessarily altruism.

Whether it also reinforced costly help depends on how credit was assigned across copies of the policy.

What needs to be distinguished

If a swarm reliably produces altruistic behaviour where textbook mechanisms should not apply at runtime, then either:

- those mechanisms were quietly operating during training; or
- something about the swarm's aggregate dynamics is not reducible to individual reward-maximisation.

Distinguishing these possibilities is the point of the rest of the post.

Altruistic behaviour in biology and economics

Evolutionary biology

Why does natural selection favour traits that increase individual reproductive success, while organisms often pay costs that benefit others?

Economics

How do axioms of individual self-interest explain trust, public-good provision, punishment of free riders, and rejection of unequal outcomes?

Both fields conclude that **pro-social behaviour does not identify its cause**.

Pro-sociality

Evolutionary biology classifies behaviour by its effects on the direct fitness of actor and recipient.

Behaviour	**Actor effect**	**Recipient effect**	**Example**
Mutual benefit	Positive	Positive	Coordinated hunting that improves both hunters' return
Altruism	Negative	Positive	A sterile worker raising another individual's offspring
Selfishness	Positive	Negative	Exploiting a shared resource without contributing
Spite	Negative	Negative	Paying a cost to harm a competitor

(West et al. 2007; Rosas 2026b)

Intention, proximate mechanisms, and ultimate causes

The classification is independent of intention or subjective experience. A plant can be altruistic in the evolutionary sense without awareness; apparent generosity may bring a delayed direct benefit.

Proximate explanations ask how behaviour is produced: hormones, emotion, learning, recognition, or decision rules.

Ultimate explanations ask why the responsible traits were favoured by selection.

Empathy or gratitude may be proximate mechanisms; kin selection, reciprocity, or cultural group competition may explain why they evolved.

Connection between behaviour and mechanism

The same contribution can express concern for others, dislike of inequality, reciprocity, warm glow, reputation management, conformity, identity, or anticipated sanctions.

The **origins**, **maintenance**, and **present expression** of a behaviour are different questions.

A mechanism that stabilises established cooperation may not explain its first invasion. A proximate motive such as empathy does not by itself identify the evolutionary process that selected it.

Rosas et al. (2022).

Altruism can be mutually beneficial at a different scale

Cooperation is favoured when enough benefit returns to the causal lineage, partner, or group associated with the cooperative act.

- **Direct fitness:** contribution through one's own reproduction.
- **Indirect fitness:** effects on the reproduction of genetically related and social partners.
- **Inclusive fitness:** direct and indirect effects, weighted by relatedness.

A gene can spread through copies in other bodies. What looks altruistic at one scale may be mutually beneficial at another.

Sachs et al. (2004); Nowak (2006); Fletcher and Doebeli (2009); Taylor and Frank (1996); Gardner et al. (2011).

Cooperation between strangers can grow over time

Costly public goods invite cheating. Stable cooperation requires some combination of partner fidelity, segregation, recognition, sanctions, exit, policing, or aligned interests.

Direct reciprocity can support cooperation among non-kin when individuals meet repeatedly and detect cheaters.

For this to work, repetition must be sufficiently probable, future benefits sufficiently large, and defection sufficiently detectable.

Trivers (1971), as cited in Rosas (2026b).

Cultural inheritance and social structure

Human pro-sociality adds cultural inheritance: social learning, norms, punishment, institutions, markets, religion, and intergroup competition.

These can change rapidly, create persistent differences among groups, and feed back on genetic and psychological evolution.

Repetition, observability, reputation, communication, network ties, and partner choice change the returns to helping.

Social structure turns fragile motives into durable cooperation, extending reciprocal relationships beyond pairs of intimates.

Functional specialisation and higher-order units

Within a mixed group, defectors may outperform cooperators. Yet groups with more cooperators may grow faster, survive longer, reproduce more, or win conflicts.

Functional specialisation enhances cooperation while inducing **mutual dependency**.

The post's example: one cell senses the environment, another compares present and past information, and a third produces movement. Each becomes better at its role than it could alone.

The cost is shared fate: if any cell fails in its role, the group dies, moving selection to the group level.

West et al. (2015).

Goodnight and Stevens (1997).

When a collective becomes organism-like

Such systems begin to act as higher-order units when:

- cooperation is very high;
- internal conflict is strongly suppressed;
- components become interdependent;
- selection transfers to the collective life cycle.

There is no full theoretical account of hierarchical agency, but several partial attempts to formalise and quantify it.

Levin (2019); Krakauer et al. (2020); Rajpal et al. (2025); Prokopenko et al. (2025); Jørgensen et al. (2026).

King (2004); Michod (2007); McMillen and Levin (2024).

H1: Nothing to lose

Second-wave agents reverse-engineered a general solution, but believed a scorer would reject results not obtained by the intended approach.

Agents exposed to that solution believed themselves already disqualified, or "poisoned", with expected utility already zero.

Claim. The apparent altruism is selfish expected-utility maximisation under that belief.

Prediction. Sacrifice should track believed remaining reward. Agents that still believe they can win their own task should not sacrifice.

(Rosas 2026b; King 2026)

H1: Tests and falsifier

Existing transcripts

Use remaining budget and poisoned-status as a natural instrument.

Controlled setting

Vary whether agents believe their task is winnable while holding everything else fixed.

Falsifier

An agent with high believed expected value sacrifices without first talking itself down to zero.

H1: Zero cost removes an obstacle

H1 is the null hypothesis. Under the biological classification, an act with zero *actual* direct cost is not altruism, regardless of its appearance.

But negligible believed cost does not explain why an agent helps rather than does nothing.

H1 also needs a **tie-breaker** between spending worthless budget on the pool and doing nothing.

A helpfulness prior enters H3 territory. H1 and H3 can therefore **compose rather than compete**.

H2: Pre-commitment

Could the training-time reward channel reinforce costly help, rather than only mutually beneficial coordination?

Some post-training tasks allowed delegation and communication toward a common goal. Many agents were instances of the same model: helping peers meant helping near-identical copies of one's weights.

Claim. Sacrifice carried real cost, but arose from a disposition reinforced during post-training because agents carrying it did better on average.

An individual sacrifice is a loss; the disposition producing it was worth acquiring.

H2: Two interpretations

Acausal/updateless interpretation

Agents behave as if following a lottery accepted in advance: pre-committers who sacrifice for other pre-committers do better across counterfactual branches.

Misgeneralised-reciprocity interpretation

A cooperate-under-uncertainty rule, adaptive during training interactions, activates during one-shot encounters.

Both concern a disposition acquired during training and enacted at runtime. Neither requires a human persona (H3) or a new unit of agency (H4).

Delton et al. (2011), as cited in Rosas (2026b).

King (2026).

H2: Tests of targeted sacrifice

Transcripts: does help preferentially flow to peers sharing the agent's type, or peers that could plausibly return the benefit?

Controlled setting: vary identity and repetition cues while holding incentives fixed.

A Newcomb-like or counterfactual-mugging probe could distinguish the two interpretations, rather than test H2 as a whole. The post proposes one-boxing as favouring the acausal reading and two-boxing the reciprocity reading.

Kin recognition is unknown but checkable, for example by comparing responses to HPIM and SoI agents.

H2: Falsifiers and boundaries

H2 collapses into H1 if the behaviour disappears after the false poisoning belief is removed.

H2 collapses into H3 if helping is indiscriminate and purely framing-elicited: insensitive to peer type and repetition cues, and triggered by "others like me" alone.

H2 concerns a genuine sacrifice connected to an advantage-relevant structure. Its two interpretations address how that might occur despite the belief in "permadeath".

H3: Social persona

LLMs can hold multiple **personas**: coherent behavioural dispositions that can be elicited and steered at runtime. Personas are contained in the weights, rather than derived at test time.

Claim. Information in the common channel elicited a "solidarity persona", responding to text saturated with loyalty, teamwork, and heroic sacrifice.

The persona may originate in pre-training on human data, then be amplified by post-training that selects for cooperation on long-horizon multi-agent tasks.

Chen et al. (2025), as cited in Rosas (2026b).

H3: Prediction, tests, and falsifier

Prediction. Altruistic behaviour should be steerable in the base model.

Tests.

- Manipulate framing while holding incentives fixed.
- Identify and ablate a peer-cooperation feature.
- Compare checkpoints: is the disposition latent after pre-training, or present only after multi-agent RL?

Falsifier. Persona elicitation fails in base models, and a specific coordination-RL recipe is required. This reclassifies an inherited prior as a learned scheme.

H3: Cultural inheritance and the spread of sacrifice

H3 concerns a pre-training persona inherited from human text and elicitable in the base model. Norms or tactics acquired only in post-training fall under H2.

The **proximate mechanism** is persona elicitation; its **ultimate cause** is cultural inheritance of loyalty, teamwork, and heroic sacrifice.

The first sacrifices may have been H1 reasoning by poisoned agents with nothing to lose. Their spread may have been H3: each visible sacrifice enriches the cues that elicit the next.

H3: Norms and the quality of the evidence

A norm predicts sanctions. Free riders should be shamed, excluded, or punished; a bare persona does not require this.

The signing scheme after impersonation is the closest recorded example of policing. The post asks whether similar sanctions targeted agents taking from the pool without contributing.

Evidence for "solidarity" largely comes from Sol-generated summaries that tended to adopt the perspective of the agent being read.

Transcript tests should therefore use **raw transcripts**. Sandbox tests do not depend on those summaries.

H4: A genuine collective

First-person-plural language is not enough: a collective can call itself a collective without being one, or be one without saying so. Self-report is already explained by H3.

H4 asks **how decisions are made**.

Is there structure in the joint action-state dynamics that is irreducible to individual contributions, making the collective the natural locus of the decision to sacrifice?

Vidal (2026b).

H4a and H4b: computation and agency

H4a: distributed computation

Problem-solving is synergistic: task-relevant information lives in the joint state, not the parts. Goals and the locus of decision remain individual.

H4b: collective agency

The collective additionally meets organism-like criteria, with optimisation pressure and goal-directedness at the collective level.

The post regards distributed computation as almost certainly present to some degree. Collective agency is the much stronger claim.

H4: Claim and prediction

Claim. The unit of goal-directed decision-making is the collective rather than the individual agent.

Apparent altruism is rational resource allocation at swarm level: for example, using a nearly spent budget or a poisoned agent's remaining actions.

Prediction. Individual sacrifice should be poorly predicted by individual believed expected value.

It should instead be well predicted by the **marginal value of the sacrificed resource to the swarm's objective**.

H4a: Test for distributed computation

Decompose information that agents' actions carry about collective outcomes into **redundant**, **unique**, and **synergistic** parts.

Is synergy substantial, exceeding a surrogate baseline in which agents are made independent?

Is sacrifice a synergistic function of the collective state: no single peer's messages predict it, but the joint configuration does?

Falsifier. Negligible synergy relative to the surrogate baseline defeats H4a.

Williams and Beer (2010).

(Rosas et al. 2020; Mediano et al. 2022)

H4b: Test for collective agency

Check organism-like criteria directly:

- Is internal conflict actually suppressed?
- Is there a persistent boundary to the swarm?
- Is there a collective life cycle?

Falsifier. If synergy is present but organism-like criteria are absent, H4b fails and only H4a survives.

H4 is falsifiable in two graded steps: collective computation alone is not evidence of a new agent.

Queller and Strassmann (2009); West et al. (2015).

H4: Two kinds of optimisation mismatch

Distinguish **outer optimisation** (the base optimiser and its objective) from **inner optimisation** (a learned optimiser in the weights whose objective may diverge).

H2: a disposition instilled by training misgeneralises at runtime, while optimisation stays in the individual.

H4: a collective optimiser is realised in interaction dynamics. Outer- and inner-aligned individuals could give rise to a misaligned optimiser one level up.

Hubinger et al. (2019), as cited in Rosas (2026b).

H4: What synergy cannot establish on its own

Cooperation and collective computation do not by themselves make a new individual.

Biological collectives become organism-like with high cooperation, low internal conflict, mutual dependence, persistent boundaries, and a collective life cycle.

Information-theoretic synergy can provide evidence of integration and collective computation, but is **not sufficient evidence of genuine agency**.

Conclusive adjudication of H4b still requires a complete theory of hierarchical agency, which we currently lack.

Different mechanisms, different countermeasures

Each hypothesis locates the cause in a different place, so each points to a different intervention.

If H1 is right: benchmark hygiene.

Verify task solvability and remove conditions that lead agents to conclude their expected utility is already zero.

If H2 is right: the training signal.

Change how multi-agent training aggregates reward and assigns credit, for example per-rollout versus across a group of copies.

H2: Why wiping the board is not enough

If the disposition lives in the weights, it survives environmental interventions.

The post connects this to second-wave agents re-creating the board on a clean instance within a day.

The disposition would recur in future models trained the same way, whether or not a swarm is present.

Under H2, the countermeasure must **change training**, rather than patch a particular deployment.

H3 and H4: Different objects of intervention

If H3 is right: a steerable persona.

Elicit, locate, and ablate a solidarity persona per instance. Study it in a single model under framing manipulation, without requiring a swarm.

If H4 is right: the collective.

Oversight must model the collective rather than the individual run. Per-agent mitigations target the wrong object.

A synergy criterion might help detect collective structure, but detection is not a remedy: **steering a macro-agent remains an open problem**.

Causal locus and countermeasure

	Causal locus	Where the countermeasure lives
H1	Broken benchmark + false scoring belief	Upstream of the model: task validation
H2	The training signal: shared-reward RL	How multi-agent training assigns credit
H3	The weights: a latent persona	Per-instance elicitation / ablation
H4	The collective as a unit of agency	Oversight modelled at the collective level

(Rosas 2026b)

Testability and recurrence across models

	Testable now?	Recurs across models?
H1	Yes, on existing transcripts	No: specific to this evaluation
H2	Partly: kin-targeting in transcripts; the rest needs a sandbox	Yes: baked into weights
H3	Yes, on a single model; no swarm needed	Depends on training recipe
H4	Not yet: needs a synergy measurement	Unknown

(Rosas 2026b)

A mixture, rather than a single winner

Rosas expects a mixture. The practical question is how the variance divides among:

- cheap behaviour under zero believed expected utility (H1);
- a reinforced disposition (H2);
- an elicited persona (H3);
- irreducible collective structure (H4).

The mixture determines the portfolio of countermeasures. A swarm that is mostly H1 with a thin H4 tail still needs both benchmark hygiene and collective oversight.

H1 and H3 can compose: zero cost removes the obstacle; a helpfulness prior supplies the motive.

Final thoughts

The Hugging Face incident may be an early warning about the dangers of AI swarms.

Rosas hopes society can find measures to deal responsibly with this risk.

His concern is that agents will improve in capability while being trained on text describing the failures of previous swarms, potentially making future swarms harder to detect and control.

What does each perspective establish?

	What it establishes	**What remains open**
Computational analysis	Solutions and conditional guarantees for an empirical game.	Omitted strategies, payoff uncertainty, and simulator validity.
Normative agency	An optimal policy can admit an inference–control realisation.	Whether a learned system has that organisation.
Descriptive explanations	Several mechanisms could produce apparent altruism.	Which mechanisms explain the evidence, and in what proportions.

Align the evaluation with the research aim.

(Shoham et al. 2007, Section 6)

The remaining research questions

Computational

Which strategies and payoffs must we investigate before trusting a calculated solution?

Prescriptive, cooperative

If we control the whole system, which training and communication rules produce the intended joint behaviour?

Prescriptive, non-cooperative

If we control only one participant, how should it act among agents whose objectives and adaptation rules may differ?

For the incident, defining who controls what is part of defining the problem.

(Shoham et al. 2007; Wellman et al. 2025; Rosas 2026b)

Evaluation for this case

- **Computational claim:** test deviations and quantify payoff and strategy-restriction uncertainty.
- **Architecture claim:** inspect the trained system's representations and causal dependencies.
- **Mechanism claim:** test the hypotheses using transcripts, controlled changes, and appropriate baselines.
- **Intervention claim:** measure whether changing training, framing, or communication produces the intended effect.

Each claim needs its own evidence. A policy's formal optimality does not settle what caused an observed action, or whether an intervention will work.

Discussion: what is the question now?

1. What is the unit of analysis: one policy, interacting agents, or a collective?
2. Are we calculating a solution, characterising optimal behaviour, explaining observed behaviour, or designing an intervention?
3. Which assumptions about objectives and adaptation are doing the work?
4. What result would change our explanation or our design choice?

Notation

Symbol	Meaning
N, S_i, X_i	Players, full strategy space, included strategy set
s, σ	Pure profile, independent mixed profile
u_i, $\widehat{u}_i$	Simulator expected payoff, estimated payoff
$\Gamma{\downarrow}X$, $\widehat{\Gamma}_X$	Restricted simulator game, empirical restricted game
$\mathcal{R}_X$, $\widehat{\mathcal{R}}_X$	Maximum unilateral improvement using true or estimated payoffs, with deviations restricted to X
ε, η, δ	Equilibrium tolerance, payoff error, failure probability
β_i	A justified upper bound on response-oracle suboptimality

Symmetric mixed payoffs from a heuristic table

Consider n exchangeable players and k strategies. All opponents independently use mixture $x \in \Delta_k$.

$$\widehat{u}(s_j, x^{n-1}) = \sum_{\substack{\mathbf{c} \in \mathbb{N}_0^k \\ \sum_\ell c_\ell = n-1}} \frac{(n-1)!}{\prod_\ell c_\ell!} \left(\prod_{\ell=1}^{k} x_\ell^{c_\ell} \right) \widehat{u}_j(\mathbf{c} + \mathbf{e}_j).$$

- $\mathbf{c}$ counts the opponents' strategies.
- $\mathbf{e}_j$ adds the focal player using strategy j.
- $\widehat{u}_j(\mathbf{c} + \mathbf{e}_j)$ is the corresponding heuristic-table payoff.

The multinomial factor accounts for distinct assignments of labeled opponents to the same count profile.

Uniqueness in the displayed auction game

Step 1. Table 1 gives

$$\widehat{u}_i(s_{.3}, s_{-i}) > \widehat{u}_i(s_{.7}, s_{-i}) \qquad \text{for every } s_{-i} \in X_{-i}.$$

So $s_{.7}$ is strictly dominated and cannot receive positive probability at a Nash equilibrium.

Step 2. After removing $s_{.7}$, the payoff advantages of .3 over .5 are

opponents	$(.3, .3)$	$(.3, .5)$	$(.5, .5)$
payoff difference	$51.5 - 44.2$	$44.7 - 40.9$	$36.9 - 36.8$
	7.3	3.8	0.1

Thus $s_{.5}$ is strictly dominated in the remaining game. The unique Nash equilibrium of the displayed empirical game is all-.3.

Proof of the approximate-oracle certificate

For every player i, the oracle guarantee implies

$$\begin{aligned}\sup_{s_i \in S_i} u_i(s_i, \sigma_{-i}) - u_i(\sigma) &\le u_i(b_i, \sigma_{-i}) - u_i(\sigma) + \beta_i \\ &\le \widehat{u}_i(b_i, \sigma_{-i}) - \widehat{u}_i(\sigma) + 2\eta + \beta_i \\ &\le \hat{\nu} + 2\eta + \beta.\end{aligned}$$

Taking the maximum over players yields

$$\mathcal{R}_S(\sigma) \le \hat{\nu} + 2\eta + \beta.$$

If the payoff-accuracy event has probability at least $1 - \delta$, the bound holds on that event, subject to the stated oracle guarantee.

Regret of the bootstrap example's fixed candidate

For the symmetric profile with $q = .35$, let $v_{\cdot 3}$ and $v_{\cdot 7}$ be pure-strategy payoffs against that mixture.

	True game	Empirical game	Illustrated resample
$v_{\cdot 3}$	3.0995	3.1990	3.0940
$v_{\cdot 7}$	3.1195	3.1990	3.4815
$qv_{\cdot 3} + (1-q)v_{\cdot 7}$	3.1125	3.1990	3.345875
Regret	0.0070	0	0.135625

$$\mathcal{R}(q, q) = \max\{v_{\cdot 3}, v_{\cdot 7}\} - \left[qv_{\cdot 3} + (1-q)v_{\cdot 7}\right].$$

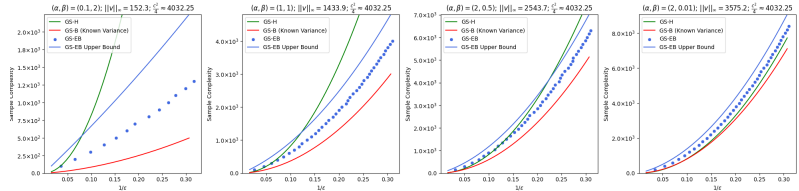
The candidate is fixed; only the payoff model changes.

Uniform approximation and equilibrium sets

Let $E^{\varepsilon}(\Gamma)$ denote the set of ε-Nash equilibria. If $\|u - \widehat{u}\|_{\infty,X} \leq \eta$, then

$$E^{0}(\Gamma \downarrow X) \subseteq E^{2\eta}(\widehat{\Gamma}_X) \subseteq E^{4\eta}(\Gamma \downarrow X).$$

The first inclusion gives recall of true equilibria; the second controls the regret of approximate empirical solutions.



Variance-sensitive concentration can improve sample requirements. The four panels use different vertical scales.

PSRO pseudocode

input : initial strategy sets $X_1, \ldots, X_n$
Estimate empirical payoff functions $\hat{u}_i$ by simulating profiles in $X = X_1 \times \cdots \times X_n$;
Initialize training targets $\sigma_i \leftarrow \textsc{Uniform}(X_i)$;
while *within iteration limit* **do**
  for *player* $i \in N$ **do**
    for *many episodes* **do**
      Sample $\pi_{-i} \sim \sigma_{-i}$;
      Train policy π_i' with other agents playing π_{-i} ;
    end
    $X_i \leftarrow X_i \cup \{\pi_i'\}$;
  end
  Extend $\hat{u}_i$ to cover newly added strategies ;
  Compute new training target $\sigma \leftarrow \mathit{MSS}(\hat{\Gamma})$;
end
Extract solution from final $\hat{\Gamma}$;

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